

CARDINHAM BODMIN, United Kingdom

WMO: 038230

Lat: 50.5022N

Lon: 4.6669W

Elev: 199

StdP: 98.96

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-0.8	-5.2	2.5	-0.2	-3.9	2.8	0.5	14.9	7.5	13.6	7.2	5.0	40	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	22.8	17.7	21.1	16.6	19.6	16.0	18.5	21.6	17.5	19.9	16.7	18.6	4.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.7	19.3	16.6	12.1	18.3	16.0	11.6	17.5	52.9	21.6	49.9	19.9	47.6	18.7	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.3	10.2	-3.6	25.5	2.0	1.6	-5.0	26.7	-6.2	27.6	-7.3	28.5	-8.7	29.7		
(5)			-4.1	20.4	1.9	1.2	-5.4	21.3	-6.5	22.0	-7.6	22.7	-8.9	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	5.6	5.6	6.8	8.4	11.1	13.5	15.4	15.3	13.7	11.3	8.3	6.2		
	DBStd	4.34	2.84	2.70	2.53	2.53	2.55	2.31	2.14	1.78	1.88	2.23	2.49	2.94		
	HDD10.0	636	138	125	103	59	16	1	0	0	0	12	62	120		
	HDD18.3	3009	396	358	359	297	224	146	96	98	139	218	302	376		
	CDD10.0	679	1	1	3	12	50	107	167	163	112	53	9	2		
	CDD18.3	9	0	0	0	0	0	2	4	2	1	0	0	0		
	CDH23.3	39	0	0	0	0	1	9	24	5	1	0	0	0		
	CDH26.7	2	0	0	0	0	0	1	1	0	0	0	0	0		
(14) Wind	WSAvg	5.1	6.0	5.8	5.6	5.2	4.9	4.5	4.2	4.3	4.6	5.2	5.3	5.8		
(15) Precipitation	PrecAvg	1207	134	105	74	73	71	71	83	91	83	132	140	134		
	PrecMax	1543	233	236	129	165	146	160	244	175	161	264	266	300		
	PrecMin	959	21	24	23	17	8	9	11	17	14	49	67	40		
	PrecStd	170	51	52	31	41	37	41	50	44	40	49	53	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	11.8	15.2	19.5	21.9	24.7	25.7	23.9	21.7	17.8	14.4	12.7		
		MCWB	11.4	10.5	10.6	13.0	16.2	18.3	18.8	19.5	17.4	15.1	13.7	12.1		
	2%	DB	11.1	10.9	12.7	16.1	19.3	21.5	23.5	21.2	19.5	16.0	13.3	12.1		
		MCWB	10.7	10.0	9.5	11.8	14.7	16.6	18.1	17.6	16.3	14.3	12.7	11.7		
	5%	DB	10.2	10.0	11.2	13.8	17.2	19.6	21.4	19.6	18.1	15.2	12.6	11.4		
		MCWB	9.7	9.2	9.3	10.4	13.6	15.5	16.9	16.6	15.4	14.0	12.0	10.9		
	10%	DB	9.5	9.3	10.2	12.2	15.4	17.8	19.3	18.3	17.0	14.5	12.0	10.5		
		MCWB	9.0	8.5	8.9	9.6	12.6	14.6	16.0	15.9	14.8	13.5	11.3	9.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.3	11.6	14.2	17.2	18.9	19.7	20.0	18.3	16.1	14.1	12.5		
		MCDB	11.7	11.6	13.2	17.7	20.2	24.1	24.3	22.8	20.1	16.9	14.3	12.6		
	2%	WB	10.8	10.2	10.6	12.2	15.4	17.1	18.5	18.3	17.1	15.1	13.0	11.8		
		MCDB	11.0	10.6	11.8	15.2	18.3	20.4	22.4	20.2	18.6	15.6	13.2	12.0		
	5%	WB	9.8	9.5	9.9	11.1	14.2	16.0	17.4	17.3	16.2	14.4	12.2	11.1		
		MCDB	10.1	9.9	10.7	13.0	16.6	18.8	20.4	18.8	17.4	14.9	12.5	11.4		
	10%	WB	9.1	8.8	9.2	10.2	13.0	15.0	16.5	16.6	15.5	13.8	11.4	10.1		
		MCDB	9.4	9.2	10.0	11.8	14.9	17.1	18.7	17.8	16.5	14.3	11.9	10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.9	5.6	6.6	6.9	6.9	6.7	6.3	6.3	5.2	4.7	4.5		
		MCDBR	4.6	4.9	6.7	8.8	9.2	9.7	10.1	7.9	7.6	5.4	4.5	4.5		
	5% WB	MCWBR	4.8	4.3	4.8	5.8	6.1	6.3	6.0	5.1	5.1	4.2	4.2	4.8		
		MCWBR	4.5	4.6	5.6	7.5	8.3	8.6	8.6	6.8	6.5	4.8	4.1	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.339	0.365	0.374	0.381	0.369	0.372	0.358	0.354	0.347	0.325	0.317		
		taud	2.373	2.402	2.341	2.339	2.354	2.412	2.409	2.467	2.468	2.460	2.422	2.337		
	Ebn at Noon		697	793	834	868	876	888	878	876	840	774	696	641		
		Edh at Noon	65	81	102	114	117	111	110	99	89	75	61	58		
(44) All-Sky Solar Radiation	RadAvg		0.80	1.59	2.73	4.39	5.33	5.88	5.36	4.57	3.43	1.94	1.02	0.64		
	RadStd		0.08	0.15	0.29	0.29	0.35	0.52	0.38	0.30	0.23	0.13	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page